

Item	Component	Specification / Model	Qty	Purpose
1	ESP32-S3	ESP32-S3 DevKitC-1	1	Master controller, Wi-Fi/BLE, scheduling, data aggregation
2	ESP8266	NodeMCU Amica V2	3	Slave controllers for room-level load control
3	Energy Meter	PZEM-004T	1	Measures AC voltage, current, power, energy, power factor
4	Voltage Sensor	ZMPT101B	2	AC voltage measurement for source monitoring
5	Current Sensor	ZMCT103C / SCT-013	2	AC current measurement for appliances and sources
6	Relay Module	12V 2 Channel 10A	3	AC load switching for controlled appliances
7	Power Supply	Hi-Link HLK-PM01 220V→5V 3W	3	Regulated DC power for controllers
8	Display	1.3" I2C OLED 128x64	3	Display real-time parameters / local UI
9	Fan	5V LED Fan	2	Cooling and airflow in enclosure
10	Light	12W 220V LED	2	Indicator / lighting
11	Temperature/Humidity	DHT11 KY-015	2	Monitor environmental conditions
12	LDR Sensor	Digital LDR Module	1	Ambient light detection
13	Veroboard	9x15cm FR4	2	Prototyping & component mounting
14	Headers	2.54mm 40x1	5	Connections for modules & sensors